

1 – (MAT-S 2012-Paper 2/1-Q6) – FUNCTIONS - ROOTS

[Maximum mark: 5]

Let $f(x) = \ln x$. The graph of f is transformed into the graph of the function g by a translation of $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$, followed by a reflection in the x -axis. Find an expression for $g(x)$, giving your answer as a single logarithm.

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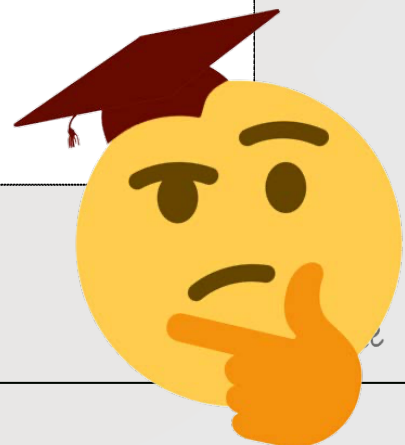
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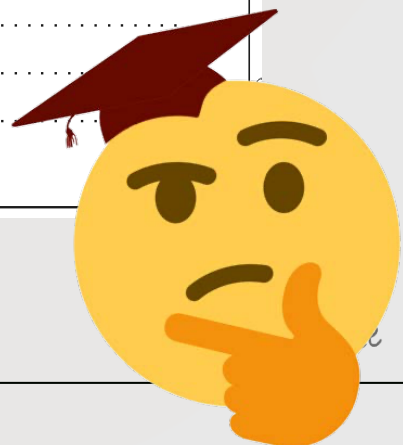


2 – (MAT-S 2014-Paper 2/2-Q7) – *FUNCTIONS - ROOTS, GRAPHS*

The function f is defined as $f(x) = -3 + \frac{1}{x-2}$, $x \neq 2$.

- (a) (i) Sketch the graph of $y = f(x)$, clearly indicating any asymptotes and axes intercepts.
(ii) Write down the equations of any asymptotes and the coordinates of any axes intercepts. [4]

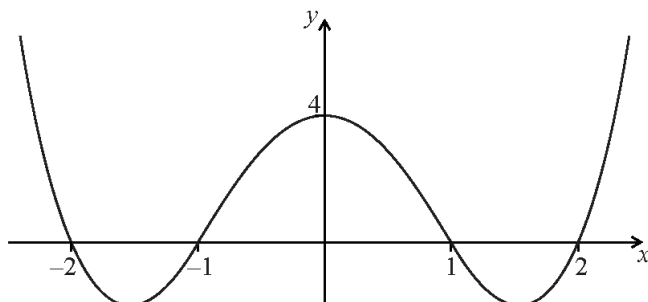
- (b) Find the inverse function f^{-1} , stating its domain. [4]



3 – (MAT-S 2014-Paper 2/1-Q12) – FUNCTIONS - ROOTS, GRAPHS

Let $f(x) = |x| - 1$.

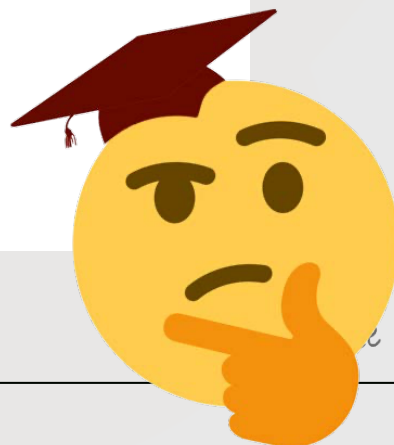
- (a) The graph of $y = g(x)$ is drawn below.



- (i) Find the value of $(f \circ g)(1)$.
- (ii) Find the value of $(f \circ g \circ g)(1)$.
- (iii) Sketch the graph of $y = (f \circ g)(x)$. [5]
- (b) (i) Sketch the graph of $y = f(x)$.
- (ii) State the zeros of f . [3]
- (c) (i) Sketch the graph of $y = (f \circ f)(x)$.
- (ii) State the zeros of $f \circ f$. [3]

- (d) Given that we can denote $\underbrace{f \circ f \circ f \circ \dots \circ f}_{n \text{ times}}$ as f^n ,

- (i) find the zeros of f^3 ;
- (ii) find the zeros of f^4 ;
- (iii) deduce the zeros of f^8 . [3]
- (e) The zeros of f^{2n} are $a_1, a_2, a_3, \dots, a_N$.
- (i) State the relation between n and N ;
- (ii) Find, and simplify, an expression for $\sum_{r=1}^N |a_r|$ in terms of n .



4 – (MAT-S 2015-Paper 2/2-Q3) – *FUNCTIONS - ROOTS, GRAPHS*

- (a) Sketch the graph of $y = (x-5)^2 - 2|x-5| - 9$, for $0 \leq x \leq 10$. [3]

- (b) Hence, or otherwise, solve the equation $(x-5)^2 - 2|x-5| - 9 = 0$. [2]

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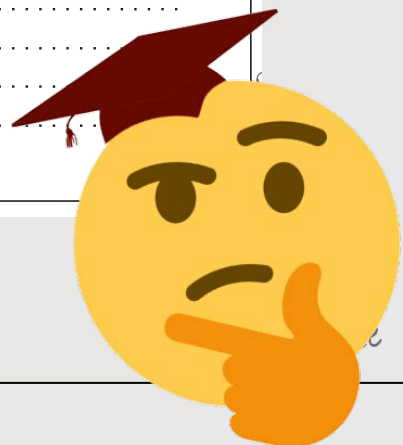
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5 – (MAT-S 2015-Paper 2/1-Q10) – *FUNCTIONS - ROOTS*

A function f is defined by $f(x) = (x+1)(x-1)(x-5)$, $x \in \mathbb{R}$.

- (a) Find the values of x for which $f(x) < |f(x)|$. [3]

A function g is defined by $g(x) = x^2 + x - 6$, $x \in \mathbb{R}$.

- (b) Find the values of x for which $g(x) < \frac{1}{g(x)}$. [7]

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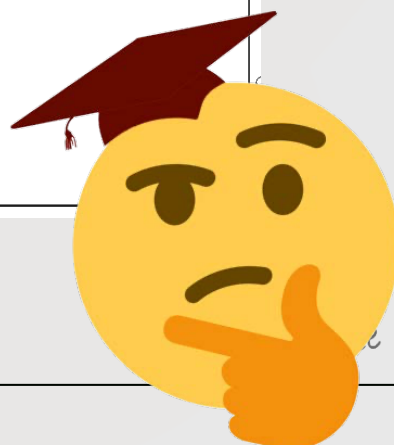
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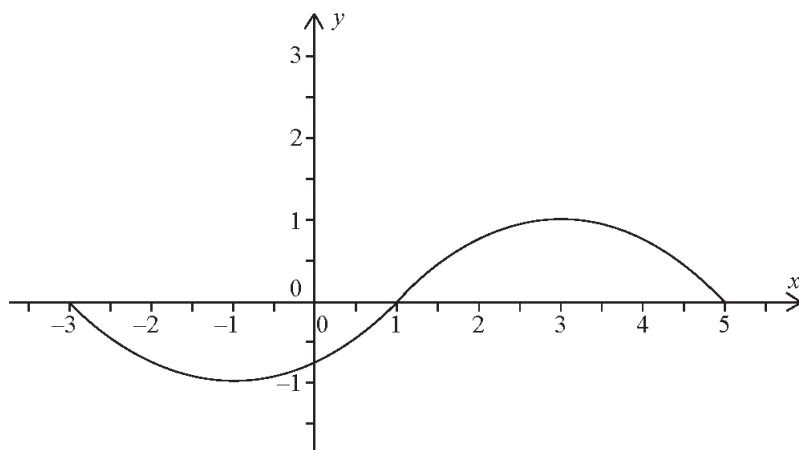
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6 – (MAT-W 2015-Paper 2/0-Q12) – FUNCTIONS - ROOTS

The following graph represents a function $y = f(x)$, where $-3 \leq x \leq 5$.

The function has a maximum at $(3, 1)$ and a minimum at $(-1, -1)$.



(a) The functions u and v are defined as $u(x) = x - 3$, $v(x) = 2x$ where $x \in \mathbb{R}$.

(i) State the range of the function $u \circ f$.

(ii) State the range of the function $u \circ v \circ f$.

(iii) Find the largest possible domain of the function $f \circ v \circ u$.

[7]

(b) (i) Explain why f does not have an inverse.

(ii) The domain of f is restricted to define a function g so that it has an inverse g^{-1} . State the largest possible domain of g .

(iii) Sketch a graph of $y = g^{-1}(x)$, showing clearly the y -intercept and stating the coordinates of the endpoints.

[6]

Consider the function defined by $h(x) = \frac{2x-5}{x+d}$, $x \neq -d$ and $d \in \mathbb{R}$.

(c) (i) Find an expression for the inverse function $h^{-1}(x)$.

(ii) Find the value of d such that h is a self-inverse function.

For this value of d , there is a function k such that $h \circ k(x) = \frac{2x}{x+1}$, $x \neq -1$.

(iii) Find $k(x)$.



7 – (MAT-S 2016-Paper 2/1-Q2) – FUNCTIONS - ROOTS, QUADRATICS

(a) Express $x^2 + 4x - 2$ in the form $(x + a)^2 + b$ where $a, b \in \mathbb{Z}$. [2]

(b) If $f(x) = x + 2$ and $(g \circ f)(x) = x^2 + 4x - 2$ write down $g(x)$. [2]

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8 – (MAT-S 2016-Paper 2/2-Q5) – FUNCTIONS - ROOTS

[Maximum mark: 6]

The function f is defined as $f(x) = \sqrt{\frac{1-x}{1+x}}$, $-1 < x \leq 1$.

Find the inverse function, f^{-1} stating its domain and range.

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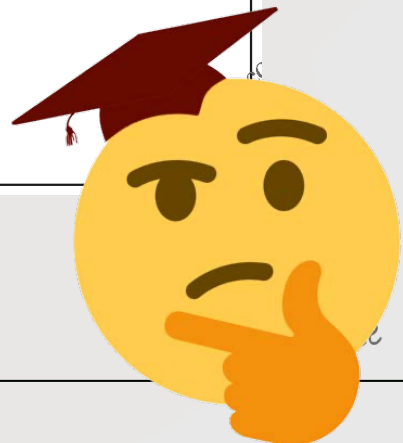
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9 – (MAT-S 2016-Paper 2/1-Q11) – FUNCTIONS - ROOTS

Let $f(x) = x^4 + 0.2x^3 - 5.8x^2 - x + 4$, $x \in \mathbb{R}$.

(a) Find the solutions of $f(x) > 0$. [3]

(b) For the curve $y = f(x)$.

(i) Find the coordinates of both local minimum points.

(ii) Find the x -coordinates of the points of inflexion. [5]

The domain of f is now restricted to $[0, a]$.

(c) (i) Write down the largest value of a for which f has an inverse. Give your answer correct to 3 significant figures.

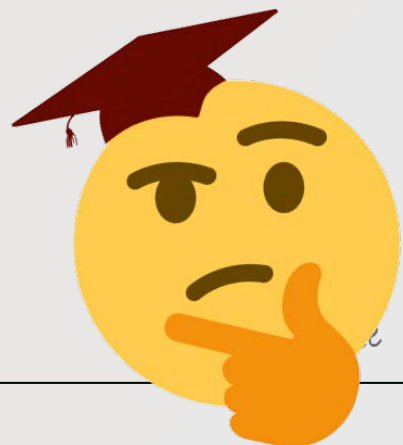
(ii) For this value of a sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same set of axes, showing clearly the coordinates of the end points of each curve.

(iii) Solve $f^{-1}(x) = 1$. [6]

Let $g(x) = 2\sin(x - 1) - 3$, $-\frac{\pi}{2} + 1 \leq x \leq \frac{\pi}{2} + 1$.

(d) (i) Find an expression for $g^{-1}(x)$, stating the domain.

(ii) Solve $(f^{-1} \circ g)(x) < 1$. [8]



10 – (MAT-W 2016-Paper 2/0-Q5) – *FUNCTIONS - ROOTS, GRAPHS*

Consider the function f defined by $f(x) = 3x \arccos(x)$ where $-1 \leq x \leq 1$.

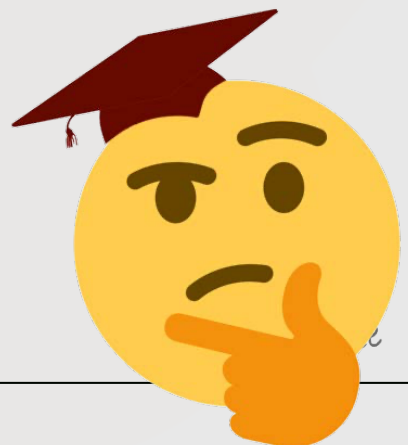
- (a) Sketch the graph of f indicating clearly any intercepts with the axes and the coordinates of any local maximum or minimum points. [3]
- (b) State the range of f . [2]
- (c) Solve the inequality $|3x \arccos(x)| > 1$. [4]



11 - (MAT-S 2017-Paper 2/2-Q11) - FUNCTIONS - ROOTS, GRAPHS

It is given that $f(x) = 3x^4 + ax^3 + bx^2 - 7x - 4$ where a and b are positive integers.

- (a) Given that $x^2 - 1$ is a factor of $f(x)$ find the value of a and the value of b . [4]
- (b) Factorize $f(x)$ into a product of linear factors. [3]
- (c) Sketch the graph of $y = f(x)$, labelling the maximum and minimum points and the x and y intercepts. [3]
- (d) Using your graph state the range of values of c for which $f(x) = c$ has exactly two distinct real roots. [3]



12 – (MAT-S 2018-Paper 2/1-Q2) – *FUNCTIONS - ROOTS*

[Maximum mark: 6]

The equation $x^2 - 5x - 7 = 0$ has roots α and β . The equation $x^2 + px + q = 0$ has roots $\alpha + 1$ and $\beta + 1$. Find the value of p and the value of q .



13 – (MAT-S 2018-Paper 2/2-Q10) – FUNCTIONS - ROOTS

Consider the expression $f(x) = \tan\left(x + \frac{\pi}{4}\right)\cot\left(\frac{\pi}{4} - x\right)$.

- (a) (i) Sketch the graph of $y = f(x)$ for $-\frac{5\pi}{8} \leq x \leq \frac{\pi}{8}$.
(ii) With reference to your graph, explain why f is a function on the given domain.
(iii) Explain why f has no inverse on the given domain.
(iv) Explain why f is not a function for $-\frac{3\pi}{4} \leq x \leq \frac{\pi}{4}$. [5]

The expression $f(x)$ can be written as $g(t)$ where $t = \tan x$.

- (b) Show that $g(t) = \left(\frac{1+t}{1-t}\right)^2$. [3]
(c) Sketch the graph of $y = g(t)$ for $t \leq 0$. Give the coordinates of any intercepts and the equations of any asymptotes. [3]
(d) Let α, β be the roots of $g(t) = k$, where $0 < k < 1$.
(i) Find α and β in terms of k .
(ii) Show that $\alpha + \beta < -2$. [7]

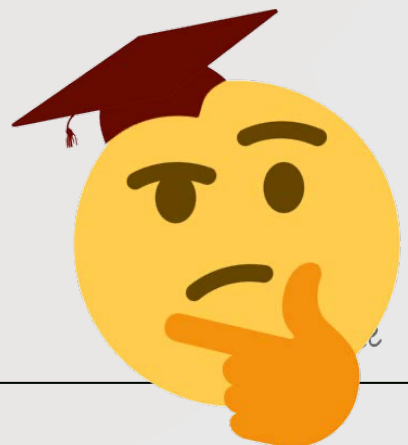


14 – (MathAA-S 2021-Paper 2/2-Q4) – ALGEBRA-FUNCTIONS - ROOTS

[Maximum mark: 6]

The functions f and g are defined for $x \in \mathbb{R}$ by $f(x) = 6x^2 - 12x + 1$ and $g(x) = -x + c$, where $c \in \mathbb{R}$.

- (a) Find the range of f . [2]
- (b) Given that $(g \circ f)(x) \leq 0$ for all $x \in \mathbb{R}$, determine the set of possible values for c . [4]



15 – (MathAA-S 2021-Paper 2/1-Q11) – ALGEBRA-FUNCTIONS - ROOTS, GRAPHS

[Maximum mark: 20]

The function f is defined by $f(x) = \frac{3x+2}{4x^2-1}$, for $x \in \mathbb{R}$, $x \neq p$, $x \neq q$.

- (a) Find the value of p and the value of q . [2]
(b) Find an expression for $f'(x)$. [3]

The graph of $y = f(x)$ has exactly one point of inflexion.

- (c) Find the x -coordinate of the point of inflexion. [2]
(d) Sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$, showing the values of any axes intercepts, the coordinates of any local maxima and local minima, and giving the equations of any asymptotes. [5]

The function g is defined by $g(x) = \frac{4x^2-1}{3x+2}$, for $x \in \mathbb{R}$, $x \neq -\frac{2}{3}$.

- (e) Find the equations of all the asymptotes on the graph of $y = g(x)$. [4]
(f) By considering the graph of $y = g(x) - f(x)$, or otherwise, solve $f(x) < g(x)$ for $x \in \mathbb{R}$. [4]



16 – (MathAA-S 2021-Paper 2/2-Q12) – ALGEBRA-FUNCTIONS - ROOTS, DIFFERENTIATION

[Maximum mark: 21]

A function f is defined by $f(x) = \arcsin\left(\frac{x^2-1}{x^2+1}\right)$, $x \in \mathbb{R}$.

(a) Show that f is an even function. [1]

(b) By considering limits, show that the graph of $y = f(x)$ has a horizontal asymptote and state its equation. [2]

(c) (i) Show that $f'(x) = \frac{2x}{\sqrt{x^2}(x^2+1)}$ for $x \in \mathbb{R}$, $x \neq 0$.

(ii) By using the expression for $f'(x)$ and the result $\sqrt{x^2} = |x|$, show that f is decreasing for $x < 0$. [9]

A function g is defined by $g(x) = \arcsin\left(\frac{x^2-1}{x^2+1}\right)$, $x \in \mathbb{R}$, $x \geq 0$.

(d) Find an expression for $g^{-1}(x)$, justifying your answer. [5]

(e) State the domain of g^{-1} . [1]

(f) Sketch the graph of $y = g^{-1}(x)$, clearly indicating any asymptotes with their equations and stating the values of any axes intercepts. [3]

